in the 16-bit era, the TurboGrafx-16 had an 8-bit CPU, a 16-bit video color encoder, and a 16-bit video display controller. However, it failed to measure up to the NES, which was still standing strong in the North American market, and faded away without much fanfare.

The first fourth generation console to make an impact was the Sega Mega Drive, the successor to Sega's Master System. While the console was sold in Japan under the Master System name in 1988, it went up as the Sega Genesis in North American markets in 1989.

Initial versions of the console had a main Motorola 68000 16-bit microprocessor, which was extended to 32-bits in later versions. The addition of

Sega Genesis

a Zilog Z80 sub-processor was to provide backward compatibility with the Master System, and also to control the audio chips. All this was bundled with 72 kB of RAM, 64 kB of video RAM, and a display that could showcase up to 61 colours at once out of 512. And like most other consoles of previous and the continuing era, ROM cartridges contained the games, and would be inserted up top into the system.

While the model didn't stand up to its main competitors particularly well in Japan, it made significant inroads in Europe, Brazil and America, where it dislodged the NES for a while, till the Super Nintendo Entertainment System was released two years later, following which they were locked in the console war for a long time.

If the Genesis is looked at as a whole, a major reason for its success was its library of arcade game ports, which provided easy access to arcade